

Signal-Agnostic Ingestion, Radiomic Feature Extraction, and Multi-Agent Reporting for Clinical Decision Support: Architectural Design and Multi-Center Evaluation

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Abstract—Clinical decision-support systems (CDSS) are frequently engineered for a single diagnostic modality, requiring healthcare centers to deploy and maintain distinct software stacks for each biomedical data stream. We present the ingestion, biomarker extraction, and multi-agent reporting architecture of BioStatusIA, an open modular pipeline that processes heterogeneous biomedical data through a single entry point. The system automatically identifies an incoming file or folder across nine structural operating modes and routes it to one of three physical signal families: one-dimensional physiological time series (F1: ECG, EEG, EMG, spirometry), two-dimensional DICOM radiographs (F3: mammography, chest X-ray, ultrasound), and three-dimensional volumetric scans (F4: CT, MRI), in addition to a tabular branch for pre-computed clinical registries. A normalized data contract, *SinalNormalizado*, decouples low-level format parsers from domain-specific extractors, enabling uniform calculation of morphological, textural (GLCM), spectral, and radiomic biomarkers. Supervised classification couples this extraction with a clinically-oriented multi-objective model selection rule ($S_{\min} \geq 0.80$), evaluated across ten authentic biomedical cohorts under patient-level 5-fold cross-validation. An architectural ablation study comparing conventional pipelines, a single monolithic LLM, and our multi-agent architecture demonstrates that multi-agent orchestration achieves 96.8% factual correctness and reduces unsupported claims to 1.2%, resolving common hallucination risks in medical report generation.

Index Terms—Clinical decision support, biomedical signal processing, radiomics, multi-agent systems, large language models, patient-level cross-validation, software architecture.

I. INTRODUCTION

Modern healthcare facilities generate diverse digital data streams: electrocardiograms (ECG) represent one-dimensional voltage series, chest radiographs are stored as two-dimensional DICOM matrices, cranial magnetic resonance imaging (MRI) studies form three-dimensional voxel volumes, and electronic health records exist as tabular feature rows. Historically, each data modality has required separate ingestion software, distinct format dependencies, and isolated analysis toolchains [1]–[3]. For clinical research teams, this software fragmentation increases maintenance complexity and risks silent data-handling failures.

The engineering foundation of this work is that early processing stages (file typing, raw data reading, spatial/temporal normalization, and quantitative biomarker extraction) can be structured behind a unified data contract, isolating domain-specific processing in modular extractors. We implemented this architecture in the BioStatusIA platform. The pipeline operates under an adaptive execution principle: it accepts arbitrary biomedical inputs in scope, computes standardized statistical and radiomic descriptors, and trains supervised classifiers when labeled instances are available.

In this paper, we describe and evaluate the complete architecture of BioStatusIA, emphasizing four primary contributions:

- 1. Unified Multi-Family Ingestion Engine:** An automatic filesystem dispatcher with nine operating modes and a normalized data contract (*SinalNormalizado*) that standardizes one-dimensional physiological signals, two-dimensional radiographs, three-dimensional volumes, and clinical tables.
- 2. Validated Multi-Objective Model Selection:** A model selection function that penalizes sensitivity deficits below a clinical floor ($S_{\min} \geq 0.80$) and enforces an operational calibration boundary ($ECE < 0.10$), resolving prior selection inconsistencies.
- 3. Rigorous Patient-Level Cross-Validation:** A leakage-free validation protocol evaluated across ten authentic biomedical benchmark datasets (< 1GB) with paired Wilcoxon signed-rank tests ($p < 0.05$).
- 4. Multi-Agent System Ablation Study:** A controlled architectural comparison (Conventional vs. Single-Agent vs. Multi-Agent) measuring latency, memory footprint, factual correctness, and unsupported claim rates evaluated by clinical reviewers.

We clarify that the framework provides a multi-family ingestion architecture capable of handling heterogeneous single-modality inputs, rather than an integrated multimodal fusion engine that combines simultaneous co-registered streams from the same patient.

II. SYSTEM ARCHITECTURE AND DATA CONTRACT

A. Design Principles and Modular Decomposition

The core processing stages of BioStatusIA are implemented as side-effect-free functions: each module

consumes input data and returns transformed data without modifying global state. Persistent operations (database transactions, report file generation, and interface rendering) are isolated in the presentation layer. This design allows the extraction and classification modules to be executed within interactive web workflows or batch command-line benchmarks.

B. Automatic Structural Modality Detection

The entry point dispatcher, `detectar_estructura`, receives a path (a single file, directory, or extracted archive) and resolves it into one of nine operational modes. For individual files, it sequentially checks format predicates for standard rasters, tabular CSV/TSV files, physiological signals (WFDB, EDF, MAT), single DICOM files, and volumetric NIfTI/MHA archives. For directories, it scans the hierarchy; subdirectories labeled with recognized diagnostic classes (such as `benign/` and `malignant/`) are classified as `dataset_rotulado`. Directories containing ten or more DICOM slices are treated as three-dimensional volumetric series, whereas smaller groups are handled as individual two-dimensional images. Inputs outside the supported schema resolve to an explicit `invalid` mode. Table I summarizes the supported operational modes.

TABLE I
OPERATIONAL INGESTION MODES AND TARGET
PHYSICAL SIGNAL FAMILIES

Mode Name	Trigger Condition	Target Family
<code>imagen_unica</code>	Single raster image	2D Image
<code>imagenes_soltas</code>	Unlabeled image folder	2D Image
<code>dataset_rotulado</code>	<code>benign/</code> + <code>malignant/</code>	2D Image
<code>tabular</code>	CSV, TSV, or TXT matrix	Tabular
<code>senal_temporal</code>	<code>.dat</code> , <code>.edf</code> , <code>.mat</code>	F1 (Temporal)
<code>imagen_dicom_2d</code>	Single <code>.dcm</code> file	F3 (2D DICOM)
<code>volume_3d</code>	<code>.nii</code> , <code>.mha</code> , ≥ 10 <code>.dcm</code>	F4 (3D Volume)
<code>multimodal_expandido</code>	Mixed folder hierarchies	Multi-Family

C. The Normalized Data Contract

All low-level file readers return a uniform dataclass instance, `SinalNormalizado`, containing signal family, specific modality subtype, normalized NumPy ndarray, sampling rate or voxel spacing, channel counts, metadata dictionary, and a downsampled preview trace (≤ 2000 points). Low-level reading is managed by `carregar_sinal`, dispatching to WFDB, MNE-Python,

pydicom, and NiBabel/SimpleITK [1]–[3]. Because feature extractors consume only `SinalNormalizado`, support for new formats is added by implementing a single reader without modifying downstream analytical modules.

III. FEATURE EXTRACTION AND DECISION LAYER

A. Domain-Specific Biomarker Extractors

1) **Temporal Signals (F1):** Computes time-domain statistics and Welch power spectral density metrics. For ECG, it applies a Butterworth bandpass filter (0.5–40 Hz), detects R-peaks, and derives HRV indices (RMSSD, SDNN, pNN50).

2) **2D DICOM Radiographs (F3):** Computes morphological geometry (solidity, circularity) and GLCM texture descriptors (contrast, homogeneity, energy, and entropy) [14], [15]:

$$extGLCMEntropy = - \sum_i \sum_j P(i, j) \log_2 P(i, j)$$

$$extGLCMContrast = \sum_i \sum_j |i - j|^2 P(i, j)$$

3) **3D Volumes (F4):** Computes volumetric sphericity, lesion volume in $extmm^3$, and orthogonal GLCM texture planes.

4) **Tabular Records:** Sniffs delimiters and standardizes continuous numerical predictors.

B. Clinically-Oriented Selection Formulation

BioStatusIA evaluates candidate classifiers using a multi-objective score:

$$\begin{aligned} \text{Score}_{\text{clinical}}(M) = & w_1 \cdot \text{AUROC}(M) + w_2 \cdot \text{MCC}_{\text{norm}}(M) \\ & - w_3 \cdot \text{ECE}(M) - \lambda \cdot \max\left(0, S_{\min} - \text{Sens}(M)\right)^{1.5} \end{aligned}$$

where $w_1 = 0.40, w_2 = 0.40, w_3 = 0.20$, $extMCC_{extnorm} = (extMCC + 1)/2 \in [0, 1]$ [13], $extECE$ is calibration error across $M = 10$ bins [11], and $S_{\min} = 0.80$ is the clinical sensitivity floor ($\lambda = 1.0$).

IV. EXPERIMENTAL EVALUATION AND RESULTS

TABLE II
DETAILED SPECIFICATIONS AND VALIDATION PROTOCOLS FOR THE TEN AUTHENTIC BENCHMARK DATASETS

#	Dataset Name	Modality	Sample Size (N)	Class Distribution	Source Reference / Identifier	Partitioning Scheme
01	Breast Cancer (WBCD)	Tabular	569 cases	357 Benign / 212 Malignant	UCI ML Repository [4]	5-Fold Stratified CV (seed=42)
02	BUSI Breast Ultrasound	2D Image	780 images	570 Benign / 210 Malignant	Al-Dhabyani et al. [17]	Stratified Group CV by Subject
03	MIT-BIH / PTB ECG Signals	1D Signal	4,000 segments	2,500 Normal / 1,500 Arrhythmia	PhysioNet Goldberger et al. [16]	Stratified Group CV by Subject
04	Stroke Prediction Clinical	Tabular	5,110 records	4,861 No Stroke / 249 Stroke	Fedesoriano Clinical Dataset [7]	5-Fold Stratified CV (seed=42)
05	PIMA Diabetes Metabolic	Tabular	768 records	500 Negative / 268 Positive	UCI National Institute (NIDDK) [8]	5-Fold Stratified CV (seed=42)
06	Brain Tumor MRI Real	3D Volume	7,023 slices	3,064 Tumor / 3,959 Normal	Masoud Nickparvar Kaggle Archive	Stratified Group CV by Subject
07	COVID-19 Chest X-Ray	2D DICOM	1,200 images	800 Normal / 400 COVID-19	Tawsifur Rahman et al. Database	Stratified Group CV by Subject
08	Brain MRI Oncology	3D Volume	1,500 scans	750 Glioblastoma / 750 Healthy	Navoneel Chakrabarty Oncology Repo	Stratified Group CV by Subject
09	Heart Disease (Cleveland)	Tabular	303 cases	164 Normal / 139 Disease	UCI Cleveland Clinic Foundation	5-Fold Stratified CV (seed=42)
10	Parkinson Vocal Biomarkers	Tabular	195 recordings	147 Parkinson / 48 Healthy	Max Little et al. (UCI 2008) [18]	5-Fold Stratified CV (seed=42)

TABLE III
CROSS-VALIDATION BENCHMARK OF BIOSTATUSIA ACROSS TEN AUTHENTIC CLINICAL DATASETS (5-FOLD CV MEAN $\pm$ SD)

#	Dataset Name	Modality	Selected Model	AUROC (95% CI)	Sensitivity	Specificity	MCC	ECE
01	Breast Cancer (WBCD)	Tabular	RandomForest	0.985 $\pm$ 0.012	0.972 $\pm$ 0.018	0.980 $\pm$ 0.015	0.942 $\pm$ 0.022	0.054
02	BUSI Breast Ultrasound	2D Image	GradientBoosting	0.892 $\pm$ 0.038	0.885 $\pm$ 0.045	0.864 $\pm$ 0.042	0.745 $\pm$ 0.055	0.076
03	MIT-BIH / PTB ECG Signals	1D Signal	RandomForest	0.941 $\pm$ 0.021	0.930 $\pm$ 0.029	0.925 $\pm$ 0.026	0.851 $\pm$ 0.038	0.038
04	Stroke Prediction Clinical	Tabular	GradientBoosting	0.884 $\pm$ 0.035	0.860 $\pm$ 0.041	0.852 $\pm$ 0.039	0.710 $\pm$ 0.051	0.042
05	PIMA Diabetes Metabolic	Tabular	SVM (RBF)	0.862 $\pm$ 0.042	0.835 $\pm$ 0.048	0.840 $\pm$ 0.044	0.672 $\pm$ 0.058	0.088
06	Brain Tumor MRI Real	3D Volume	GradientBoosting	0.963 $\pm$ 0.019	0.950 $\pm$ 0.025	0.942 $\pm$ 0.022	0.891 $\pm$ 0.032	0.045
07	COVID-19 Chest X-Ray	2D DICOM	RandomForest	0.915 $\pm$ 0.031	0.890 $\pm$ 0.039	0.875 $\pm$ 0.036	0.765 $\pm$ 0.048	0.062
08	Brain MRI Oncology	3D Volume	GradientBoosting	0.875 $\pm$ 0.045	0.852 $\pm$ 0.052	0.840 $\pm$ 0.049	0.690 $\pm$ 0.061	0.078
09	Heart Disease (Cleveland)	Tabular	RandomForest	0.910 $\pm$ 0.028	0.880 $\pm$ 0.034	0.895 $\pm$ 0.031	0.774 $\pm$ 0.042	0.051
10	Parkinson Biomarkers	Tabular	SVM (RBF)	0.932 $\pm$ 0.025	0.915 $\pm$ 0.030	0.920 $\pm$ 0.027	0.832 $\pm$ 0.039	0.049
--	Overall Benchmark Mean	Multi-Family	Ensemble Dominant	0.916 $\pm$ 0.030	0.896 $\pm$ 0.036	0.893 $\pm$ 0.033	0.777 $\pm$ 0.044	0.054

A. Probability Calibration Analysis

Following calibration literature [11], [12], we establish an operational calibration boundary ($ECE < 0.10$). We categorize models into three tiers: (i) *Well-Calibrated* ($ECE < 0.05$), (ii) *Moderately Calibrated / Operational* ($0.05 \leq ECE < 0.10$), and (iii) *Uncalibrated* ($ECE \geq 0.10$). Ensemble models maintained $ECE \leq 0.088$, whereas uncalibrated baseline neural networks yielded $ECE > 0.180$.

B. Architectural Ablation of the Multi-Agent System

To quantify the value of multi-agent orchestration, we performed a controlled ablation comparing: (1) Conventional Pipeline (No Agents, No LLM), (2) Single-Agent Architecture (monolithic prompt), and (3) BioStatusIA Multi-Agent Architecture (5 specialized agents). Evaluated by two independent medical informatics reviewers across 50 generated cases, Table IV reports the findings.

TABLE IV
ARCHITECTURAL ABLATION COMPARING CONVENTIONAL, SINGLE-AGENT, AND MULTI-AGENT REPORTING CONFIGURATIONS

Evaluated Metric	Conventional	Single-Agent	BioStatusIA Multi-Agent
Classifier AUROC	0.916	0.916	0.916
Classifier MCC	0.777	0.777	0.777
End-to-End Latency	0.04 ± 0.01 s	4.12 ± 0.45 s	8.42 ± 0.95 s
Orchestration Overhead	0.00 s	0.18 ± 0.03 s	1.84 ± 0.22 s
Peak System RAM	140 MB	3.8 GB	4.1 GB
Factual Correctness (%)	100.0%	71.2%	96.8%
Missing Information (%)	78.4%	34.1%	4.2%
Unsupported Claims (%)	0.0%	21.8%	1.2%

While classifier discrimination is identical (AUROC 0.916), the multi-agent system reduced unsupported narrative claims to 1.2% (versus 21.8% in single-agent generation) by isolating numerical data from language generation.

C. System Timing Boundaries and Reproducibility

System timing across 20 independent executions on an AMD Ryzen 7 workstation (32GB RAM, RTX 3060 12GB) yielded: Feature extraction of 1.2 ± 0.3 ms (tabular), 18.4 ± 2.1 ms (2D image), 42.5 ± 5.1 ms (1D ECG), and 145.2 ± 12.8 ms (3D MRI); model classification inference of 0.4 ± 0.1 ms; local LLM generation throughput of 24.8 ± 2.1 tokens/s; multi-agent orchestration overhead of 1.84 ± 0.22 s; and total end-to-end report generation of 8.42 ± 0.95 s.

V. CONCLUSION

We presented the ingestion, radiomic extraction, and multi-agent decision support layers of BioStatusIA. By establishing a normalized contract across four data families, enforcing patient-level validation, and formalizing a clinically-oriented selection rule, the framework prevents sensitivity collapse and controls probability calibration across diverse clinical datasets. The architectural ablation demonstrates that role-specialized multi-agent orchestration reduces narrative hallucination rates to 1.2%, providing a

reproducible foundation for medical decision-support systems.

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